**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**
**Department of Artificial Intelligence and Data Science**

## ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

---

| | | | |
|---|---|---|---|
| **Course Code:** | DSA03 | **Course Title:** | Natural Language Processing |
| **CO Assessed:** | CO5 – Articulation | **Assessment:** | Assessment Tool 1 – Constraint-Based Problem Solving |
| **Weightage:** | 35% of CIA | | |
| **CO5 Statement:** | Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5) | | |
| **Student Name:** | A. Shiva Shankar Reddy | **Reg. No.:** | 192425174 |
| **Section / Batch:** | CSE AIML | **Date:** | 28/08/26 |

### Code of Conduct

*I **A. Shiva Shankar Reddy/ 192425174** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*
Signature of the Student: _____A Shiva Shankar Reddy____________________

**Instructions**

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

| Section / Topic | Focus | Question No. |
| --- | --- | --- |
| Discourse, Reference Resolution & Text Coherence | Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph. | Q1 |
| Dialog and Conversational Agents, Discourse Planning & Surface Realizations | Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output. | Q2 |
| Machine Translation, Discourse Structure, Predicate Logic & Interpretation | Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases. | Q3 |

**Annexure A – Constraint-Based Problem Solving**

---

1. **Given the following paragraph:**

"John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

**Tasks:** a) Identify all referring expressions (mentions) and possible antecedents.

b) Apply the following **constraints** to resolve the references:

- Gender and number agreement
- Recency (prefer recent antecedents)
- Semantic compatibility (e.g., animacy and action suitability) ● Coherence (maintain consistent entity chains)

c) Draw a **constraint graph** or table showing variables, domains, and which constraints eliminate possible links.

d) Provide the final resolved coreference chains and rewrite the paragraph with clear references.

e) Justify the priority order of your constraints and discuss what would happen if the recency constraint is relaxed.

2. **Conversation History:** User: “I have an important exam tomorrow but I’m not able to concentrate.”

**Required Dialog Act:** Advise + Encourage (Give advice and motivate the student).

**Constraints:**

1. Maintain **entity coherence** (reuse “exam”, “concentrate”, “you”).
2. Use **discourse relations** (Cause-Effect or Elaboration).
3. Include at least **two** of these keywords: focus, break, confident.
4. Response length: 2–3 sentences only.
5. Logical consistency (no contradictory statements).
6. Politeness and positive tone.

**Tasks:** a) Identify the relevant discourse planning steps needed before generation.

b) Generate **three** different possible responses that satisfy **all** the above constraints.

c) Evaluate the three responses and select the best one. Justify your choice using coherence and constraint satisfaction criteria.

d) Explain how violating any two constraints would affect the quality of the dialog.

3. **Source Sentence:** "The bank by the river flooded after the storm, but it was saved by quick action."

**Note:** "bank" is ambiguous (financial institution or riverbank).

**Constraints:**

1. Correct **Word Sense Disambiguation** using context.
2. Create a **predicate logic** representation (e.g., using predicates like flood(x), location(x,y)).
3. Maintain **discourse structure** (Contrast relation between the two clauses).
4. Preserve all important entities and relations (no information loss).
5. Ensure **text coherence** in the generated output.

**Tasks:** a) Resolve the ambiguous word "bank" and justify using constraints.

b) Write a predicate logic representation of the sentence.

c) Generate a good **target language sentence** (in any Indian language or simple English paraphrase) that satisfies all constraints.

d) Draw a simple **discourse tree** (RST-style) showing the rhetorical relation.

e) Discuss the advantages of using constraint-based approach over pure statistical machine translation for this sentence

# QUESTION 1 – Discourse, Reference Resolution & Text Coherence

Given paragraph:

*"John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."*

The task is to resolve the referring expressions by applying gender and number agreement, recency, semantic compatibility, and coherence, and then maintain consistent entity chains.

**1(a). Referring expressions and possible antecedents**

| Mention | Possible antecedents | Initial interpretation |
|---|---|---|
| He | John, Mary | John is compatible by gender and number. |
| She | John, Mary | Mary is compatible by gender and number. |
| it | ball, park, dog | Ball is semantically suitable for "play with it". |
| him | John, Mary | John is compatible by gender; Mary is excluded by gender. |
| they | John, Mary, dog / group | Plural reference to the people and dog in the discourse. |

**1(b). Constraint-based resolution**

- Gender and number agreement eliminates antecedents whose grammatical gender or number does not match the referring expression.
- Recency prefers a recently mentioned compatible entity, but it is not sufficient by itself.
- Semantic compatibility checks whether the antecedent can naturally participate in the described action.
- Coherence checks whether the resulting entity chain remains consistent across the paragraph.

For "He", John is selected because the expression is singular masculine and John is the compatible named person. For "She", Mary is selected because the expression is singular feminine. For "it", the ball is the strongest antecedent because a person can naturally play with a ball, while the park and dog do not fit the intended action. For "him", John remains the only gender-compatible human antecedent. For "they", the plural group refers to the discourse participants, including John and Mary and, in the broad final-event reading, the dog.

**1(c). Constraint graph / decision table**

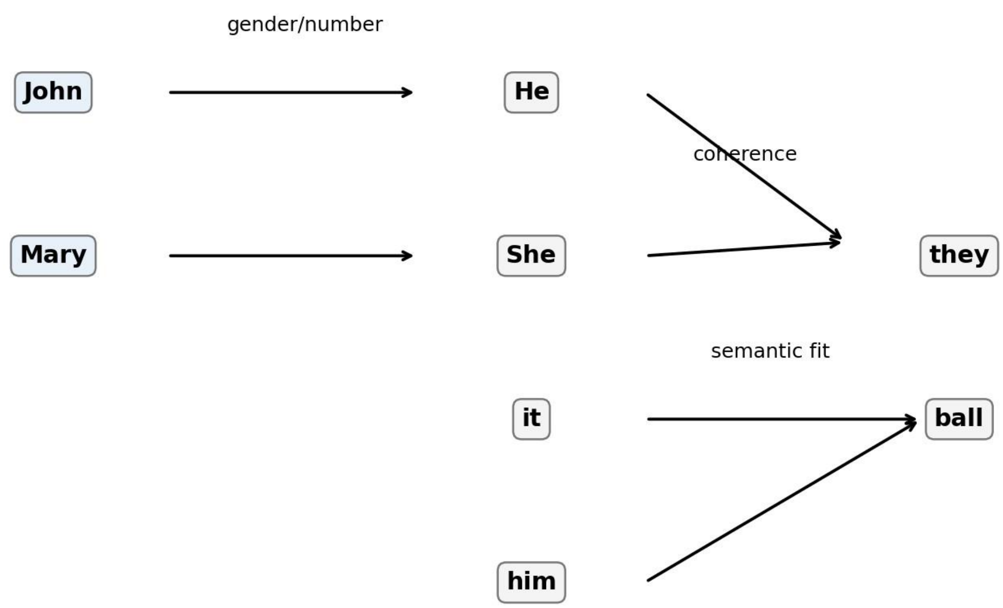


*Figure 1. Constraint-based coreference graph showing candidate links and the main resolution constraints.*

**1(d). Final coreference chains and rewritten paragraph He → John She → Mary it → the ball him → John they → John + Mary + dog**

Rewritten paragraph: “John and Mary went to the park. John brought a ball. Mary wanted to play with the ball. The dog chased John excitedly. Finally, John, Mary, and the dog all went home.”

**1(e). Priority order and effect of relaxing recency**

Recommended priority: (1) agreement, (2) semantic compatibility, (3) discourse coherence, and (4) recency as a tie-breaking preference. Agreement and semantic compatibility are stronger because they eliminate impossible or unnatural links. Coherence then ensures that the selected antecedents form a consistent discourse model. Recency is useful when several candidates remain possible.

If recency is relaxed, the system can still resolve references correctly when agreement and semantic compatibility provide strong evidence. However, ambiguous cases may retain more candidates, increasing computational effort and the possibility of selecting an older but less likely antecedent.

**Conclusion**

Constraint-based reference resolution is effective because it combines grammatical, semantic and discourse evidence instead of relying on a single heuristic. The final chains preserve entity consistency across the paragraph.

# QUESTION 2 – Dialog and Conversational Agents, Discourse Planning & Surface Realizations

Conversation history: "I have an important exam tomorrow but I'm not able to concentrate." Required dialog act: Advise + Encourage. The response must maintain entity coherence, use a Cause–Effect or Elaboration relation, contain at least two of the keywords focus, break, confident, contain 2–3 sentences, remain logically consistent, and use a polite positive tone.

**2(a). Discourse planning steps before generation**

- Identify the user state: an important exam is tomorrow and concentration is currently difficult.
- Select the required dialog acts: first provide practical advice and then encourage the user.
- Choose a discourse relation: use Cause–Effect or Elaboration so that the advice logically supports the encouragement.
- Plan the required vocabulary: include at least two of "focus", "break", and "confident".
- Check surface constraints: exactly 2–3 sentences, positive/polite tone, no contradictions, and reuse the relevant entities "exam", "concentrate", and "you".

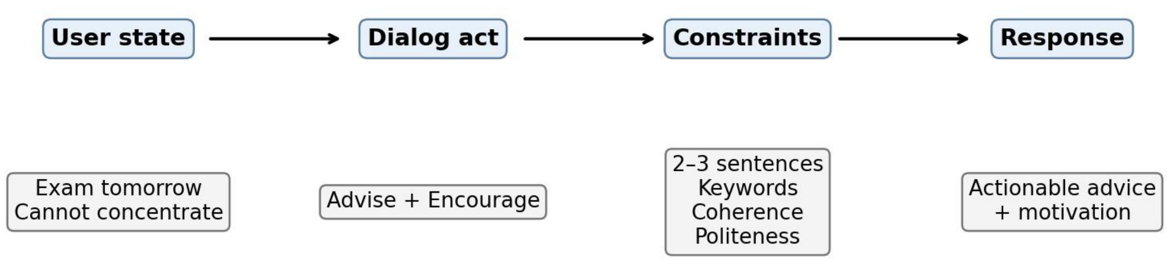


*Figure 2. Constraint-based dialogue generation pipeline.*

**2(b). Three valid constrained responses**

**Response 1**

Since your exam is tomorrow, take a short break and return with a clear focus on one topic at a time. This can help you concentrate better and feel more confident.

**Response 2**

You can improve your concentration by taking a short break and then setting a small study goal for the exam. Keep your focus on that goal, and you can feel more confident about your preparation.

**Response 3**

Because the exam is tomorrow, avoid trying to study everything at once; take a short break and study one important topic at a time. Stay focused and remind yourself that you can be confident in your preparation.

**2(c). Evaluation and best response**

| Response | 2–3 sentences | Keywords | Coherence | Overall |
|---|---|---|---|---|
| 1 | Yes | focus, break, confident | Strong cause–effect | Best |
| 2 | Yes | focus, break, confident | Strong cause–effect | Very good |
| 3 | Yes | break, focus, confident | Strong cause–effect | Very good |

Response 1 is selected as the best because it satisfies all constraints directly and naturally. It clearly connects the exam situation to a short break and focused study, contains all three useful keywords, remains within two sentences, and ends with positive encouragement.

**2(d). Effect of violating two constraints**

If the 2–3 sentence length constraint is violated, the response may become too long and less suitable for a conversational assistant, reducing clarity and user engagement. If the keyword or entity-coherence constraint is violated, the answer may fail the specified generation requirements and may appear disconnected from the user's stated problem.

**Conclusion**

Constraint-based dialogue generation ensures that the response is not only grammatically acceptable but also aligned with the required dialog act, discourse relation, tone, vocabulary and length.

# QUESTION 3 – Machine Translation, Discourse Structure, Predicate Logic & Interpretation

Source sentence: "The bank by the river flooded after the storm, but it was saved by quick action." The word "bank" is ambiguous between a financial institution and a riverbank. The solution must preserve the intended sense, represent the meaning using predicates, maintain the Contrast relation, preserve important information, and remain coherent.

**3(a). Word Sense Disambiguation**

The correct sense is riverbank. The strongest constraint is the explicit contextual phrase "by the river". A financial institution does not naturally occur "by the river" in the intended physical-event interpretation of "flooded"; a riverbank can flood directly. The predicate event "flooded" also has strong semantic compatibility with a physical riverbank.

**bank → RIVERBANK**

Constraint evidence: location context ("by the river") + event compatibility ("flooded") + discourse coherence ("it was saved by quick action"). Therefore, the financial-institution sense is eliminated. **3(b). Predicate logic representation**

**Bank(b)**

**River(r)**

**By(b,r)**

**Storm(s)**

**Flooded(b,s)**

**QuickAction(q)**

**Saved(b,q)**

A compact event representation is: $\text{Flooded}(b) \land \text{After}(\text{Flooded}(b), \text{Storm}(s)) \land \text{Saved}(b,q) \land \text{QuickAction}(q)$. The variable b denotes the riverbank, s the storm, and q the quick action. The representation preserves the key entities and event relations stated in the source sentence.

**3(c). Coherent target-language sentence / paraphrase**

Simple English paraphrase: "The riverbank near the river flooded after the storm, but quick action saved it."

This version keeps the riverbank sense of "bank", preserves the storm and flooding event, retains the contrast introduced by "but", and preserves the outcome that quick action saved the bank.

### 3(d). RST-style discourse tree

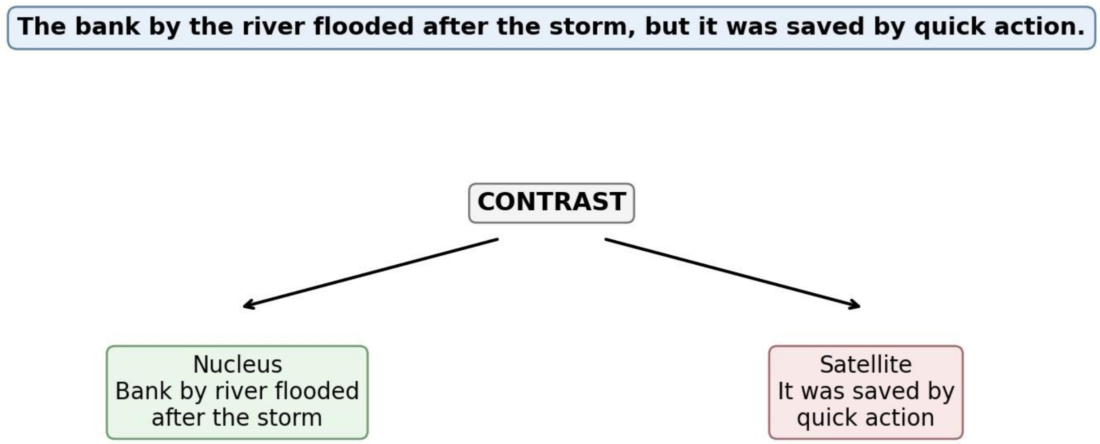


*Figure 3. RST-style discourse structure showing the Contrast relation between the flooding event and the successful rescue.*

### 3(e). Advantages of constraint-based MT over pure statistical MT

- Constraint-based translation explicitly uses contextual and semantic constraints, reducing ambiguity in words such as "bank".
- Predicate representations make important entities and relations explicit, helping prevent information loss.
- Rhetorical constraints preserve discourse relations such as Contrast, which supports coherent targetlanguage output.
- The reasoning is more interpretable because the system can explain why one word sense was selected.
- A pure statistical approach may select a frequent sense from training data even when the local context strongly indicates another sense.
- Constraint-based methods are especially useful when a sentence contains high-impact ambiguity or when preservation of specific logical relations is required.

### Conclusion

For the supplied sentence, contextual constraints correctly identify "bank" as riverbank. Predicate logic preserves the entities and events, while the RST Contrast relation maintains the discourse structure. This demonstrates how explicit constraints can improve semantic reliability and translation coherence.

The supplied assessment explicitly requires necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate-logic representations; solutions that satisfy all constraints; clear evaluation; and concise, evidence-supported answers. filecite turn1file0 L18-L23

The document therefore includes a coreference constraint graph for Q1, a constrained dialoguegeneration flow for Q2, and an RST-style discourse tree plus predicate logic for Q3. The three questions and their task structure are preserved from Annexure A. filecite turn1file0 L50-L94

**Rubric-focused note**

The solutions emphasize problem understanding, explicit constraint analysis and prioritization, feasible solutions satisfying the stated constraints, correct application of NLP concepts, and logical justification—the five assessment dimensions identified in the supplied rubric.
filecite turn1file0 L105-L197